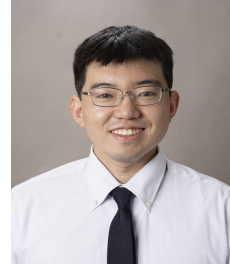


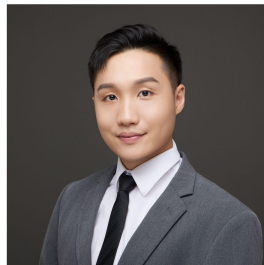
Restaurant AI Checkpoint 4



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<https://github.com/IS492-SP26/Restaurants-AI/tree/main>

web app: <https://restaurants-ai-48ke.vercel.app/>

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Recap and Final Goal



Problem Statement

Growing interest in local entrepreneurship combined with limited guidance leads to costly mistakes. AI can integrate modeling and checks to reduce risk.

Target Users



Early entrepreneurs



Established owners

User-Study Goal

- Achieve high satisfaction across usability scales
- Identify potential bottleneck or improvements to current system



Evaluation Design Overview

Quantitative Survey



System Usability Scale (SUS)

- Adopted for standardized assessment.
- From Digital Equipment Corporation
- Scale of 1 to 5 (Disagree to Agree)

10 Questions

Focus: Ease of use, Complexity, and Reliability.

Qualitative Survey



Detailed Feedback

6 Questions to capture user insights.

Key Dimensions

Product efficacy, usability, and future improvements.



Total Participants: 9

Quantitative Results (1)



Question	Average Rating (1 to 5) - Rounded
1. I think I would like to use this system frequently	4
2. I found the system unnecessarily complex	1.78
3. I thought the system was easy to use	4.33
4. I would need the support of a technical person	1.56
5. The various functions were well integrated	4.56



Quantitative Results (2)



Question	Average Rating (1 to 5) Rounded
6. I thought there was too much inconsistency	1.78
7. Most people would learn to use this quickly	4.44
8. I found the system very cumbersome to use	1.44
9. I felt very confident using the system	3.78
10. I needed to learn a lot before getting going	2.11



Qualitative Insights



Top Positive Themes

Locally Grounded Output

Champaign-specific citations (CUPHD rules, county wages, campus corridors) distinguished our tool from generic AI

Structured Beginner-Friendly Workflow

3-step flow (Basics → Ideas → Manual) was intuitive to all participants — zero assistance needed

Regulatory & Financial Anchoring

Cited real codes (Rule 750.4000, Sec. 37-5) and macro data (CPI-U, BLS wages) — verified by domain experts



Top Frustrations

Financial Personalization Gap

Budget shows % ranges but never calculates dollar amounts from user's input. No pro forma or break-even template provided.

Mock Data & Technical Labels

"Fine-tuned agents active: 0" badge and mock competitor placeholders confused non-technical users and undermined credibility.



Interpretation & Discussion

81.1

Avg SUS Score (Above 68 benchmark)

SUS Score	Percentile range	Grade	Rank
84.1 - 100	96 - 100	A+	Best imaginable
80.8 - 84.0	90 - 95	A	
78.9 - 80.7	85 - 89	A-	
77.2 - 78.8	80 - 84	B+	Excellent
74.1 - 77.1	70 - 79	B	
72.6 - 74.0	65 - 69	B-	
71.1 - 72.5	60 - 64	C+	Good

Local grounding is our key differentiator

All participant types — from compliance experts to restaurant staff — consistently distinguished our tool from generic AI by its Champaign-specific citations. This local data anchoring creates credibility that general-purpose models cannot match.

The tool succeeds as a planning accelerator, not a standalone planner

Users trust it for brainstorming and regulatory awareness , but want professional verification for financial decisions. The "AI as co-pilot" positioning is validated.

The gap from guidance to action is the primary improvement area

Users want calculated dollar figures, downloadable templates, and worked examples — not just directional percentages and advice.



Limitations & Risks



Limitations

Small sample size (n = 9)

Diverse backgrounds but insufficient for statistical generalizability. Results reflect qualitative patterns.

Ecological validity

Testing in a study setting differs from real-world restaurant launch conditions with higher stakes and time pressure.

Single geography scope

Tool is built for Champaign, IL only. Regulatory and market data don't transfer to other jurisdictions.

Risks

Hallucination risk

AI-generated regulatory/financial content may contain inaccuracies. Over-reliance without professional verification poses compliance risk.

Model unpredictability

Outputs may vary across runs for identical inputs. "Confidence: Medium" indicators were unclear to users.

Over-reliance risk

High enough in the tool can lead naive users to treat AI outputs as authoritative rather than directional.



Conclusion & Future Work



Main Takeaways

Multi-agent AI meaningfully lowers the barrier to restaurant planning.

The high usability and comprehensiveness allows access to local grounding validates structured, locally-aware AI over generic outputs.

AI as co-pilot, not replacement.

Users trust the tool for brainstorming and regulatory scaffolding but still want professional sign-off for financial and legal decisions.

Future Improvements

1

Financial personalization

Convert % ranges to dollar amounts using user's stated budget; add downloadable break-even templates

2

Real competitor data

Replace mock placeholders with actual Champaign restaurants via API integration

3

Side-by-side concept comparison

Redesign Ideas page with a comparison table highlighting key differences

4

Expand regulatory coverage

Add IL sales tax registration, Champaign liquor commission, employer compliance



**Thank you for
listening !**
